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Saveetha Institute of Medical And Technical Sciences

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Fresh Intelligent Spam and Malicious URL Detection System

CSA5111 - CRYPTOGRAPHY AND NETWORK SECURITY

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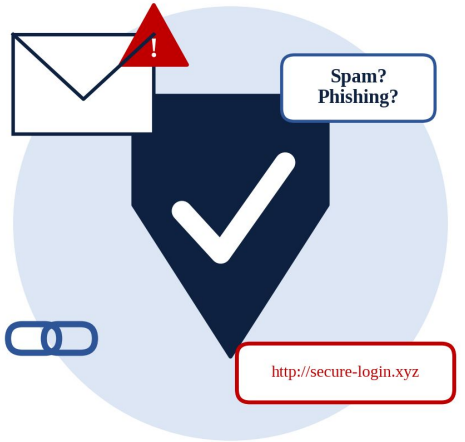
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Introduction

- Spam messages and suspicious URLs are increasing in volume, creating risks such as **phishing**, fraudulent links and malicious websites.
- Manually identifying suspicious messages and URLs is slow and difficult, because they are often made to look legitimate.
- Unsafe links and messages can mislead users into sharing information or visiting harmful sites.
- Feature extraction and **machine learning** can help classify suspicious messages and URLs automatically.
- **Risk scoring** and a **security dashboard** help users understand and review detected threats.



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Problem Statement

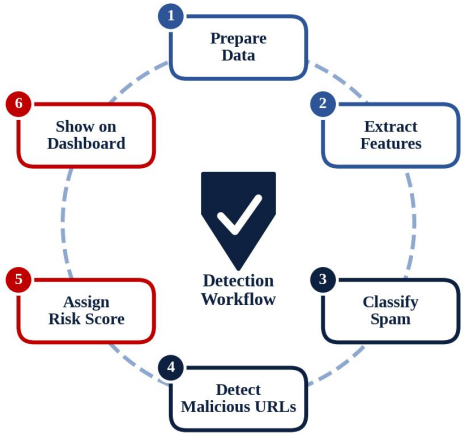
- Spam and malicious URLs are hard to separate from **legitimate content**, especially when they imitate trusted messages and websites.
- Suspicious links can **mislead users** or expose them to cybersecurity threats such as phishing.
- Relying only on **manual inspection or simple rules** has limits and may miss new or changing patterns.
- There is a need to **analyze relevant message and URL characteristics** to support detection.
- Detection results and **risk levels** must be presented in an understandable format for review.

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Objectives

The proposed system aims to:

- Prepare datasets for model development and evaluation.
- Extract useful features from spam and URL data.
- Classify messages as **spam or legitimate** using a suitable ML approach.
- Identify potentially **malicious URLs** from URL characteristics.
- Assign interpretable **risk scores or categories**.
- Present results and warnings in a **security dashboard**.
- Evaluate the system with appropriate metrics and test data.



Literature Review

S.No	Year	Reference	Related Work Summary	Tools / Tech	Data / Components	Key Finding / Limitation
1	2022	Almeida et al., ACM DocEng	Introduced the public SMS Spam Collection with baseline results.	Several ML methods	Labelled SMS (spam/ham)	Public benchmark; covers SMS only.
2	2020	Blanzieri & Bryl, Artificial Intelligence Review	Surveyed learning-based email spam filtering and its evaluation.	ML-based spam filters	Filter design, evaluation methods	Survey; no new system proposed.
3	2024	Ma et al., ACM KDD	Classified URLs using lexical and host-based features.	Statistical URL classification	Labelled URLs, lexical + host features	Reported 95–99% accuracy on their data.
4	2024	Sahoo et al., arXiv:1701.07179	Surveyed ML for malicious URL detection.	Survey of ML methods	Feature representation, algorithms	Blacklists cannot be exhaustive.

Existing System and Its Limitations

Existing System

- Manual identification and reporting of suspicious messages.
- Spam filters using predefined rules or conventional classification methods.
- URL reputation checks and blacklist-based detection of known bad links.
- Separate tools for message classification, URL checking and threat reporting.

Limitations

- New or unseen threats may not be covered by existing rules or lists.
- Poorly selected features may reduce classification effectiveness.
- Separate tools may not give one unified view of threats.
- False positives can affect legitimate messages or websites.
- A result without meaningful risk information is hard to interpret.

Proposed System and Advantages

Proposed System

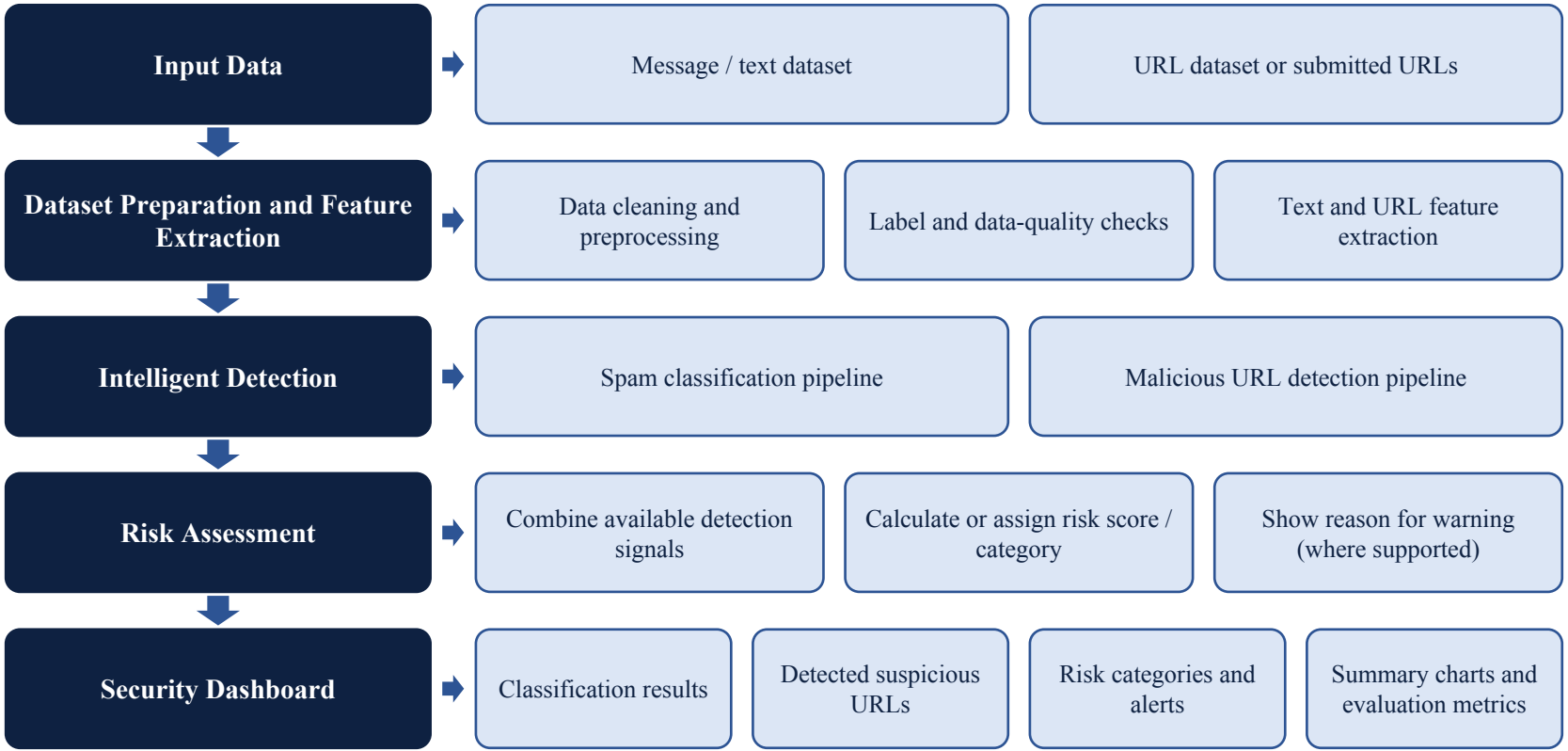
A unified system with four components: dataset preparation and feature extraction, intelligent spam classification, malicious URL detection, and risk scoring with a security dashboard. It analyzes submitted or test data and presents detection results and risk levels in one place (proposed design).

Advantages

1. Structured dataset preparation and feature extraction.
2. Automated spam classification.
3. Dedicated malicious URL analysis.
4. Risk-based interpretation of detection results.
5. A common dashboard for reviewing alerts and summaries.
6. Measurable evaluation using appropriate test data.

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System Architecture



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Modules

Module 1: Dataset Preparation and Feature Extraction

Module 2: Intelligent Spam Classification

Module 3: Malicious URL Detection

Module 4: Risk Scoring and Security Dashboard

Module 1:
Dataset Preparation and Feature Extraction

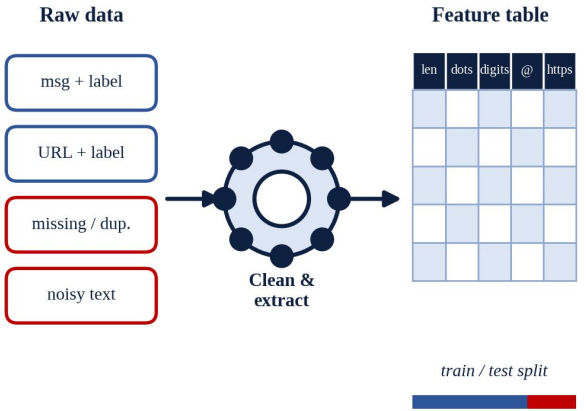
Purpose: Prepare suitable data and convert it into features for detection.

Functions

- Load the message and URL datasets used.
- Inspect fields, labels, missing values and duplicates.
- Clean and preprocess text and URL records.
- Extract text features from message content.
- Extract URL features (length, hostname structure, other lexical or structural properties).
- Prepare features for the detection pipelines.
- Document the train/test split and preprocessing steps.

Input: Available labelled or unlabelled message and URL datasets.

Output: Cleaned data and feature representations for model development or detection.



Functions describe the proposed design.

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Module 2:
Intelligent Spam Classification

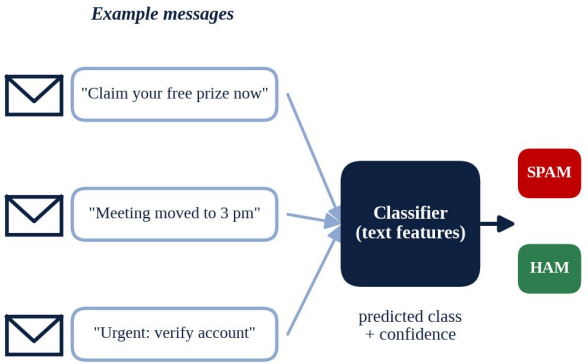
Purpose: Classify message or text records as spam or non-spam.

Functions

- Receive preprocessed text and its extracted features.
- Represent text with a suitable encoding approach.
- Apply a suitable classifier (algorithm to be finalized).
- Produce the predicted class and, if available, a confidence score.
- Evaluate performance with appropriate metrics.
- Pass results in a format the dashboard can use.

Input: Preprocessed text, extracted features and trained model (where applicable).

Output: Predicted class, plus confidence or evaluation information if supported.



Functions describe the proposed design.

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Module 3:
Malicious URL Detection

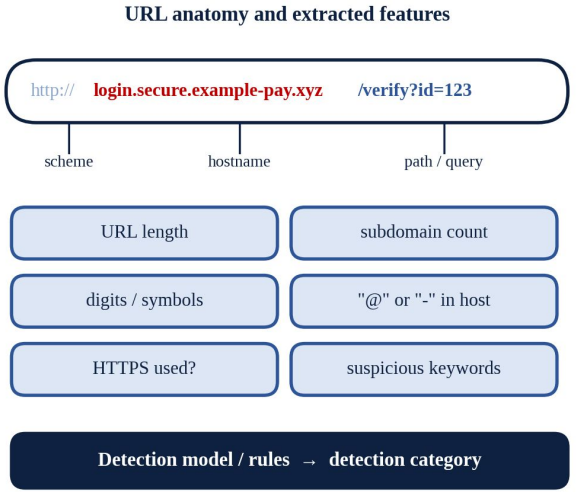
Purpose: Analyze URL characteristics and identify potentially malicious or suspicious URLs.

Functions

- Accept a URL or a record from the URL dataset.
- Validate and parse the URL where appropriate.
- Extract supported URL-based features.
- Apply the project's detection rules or trained model.
- Produce a prediction or detection category.
- Pass detection information to risk scoring.

Input: URLs, labels when available, and extracted URL features.

Output: Detection result and associated risk information, where supported.



Functions describe the proposed design.

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Module 4:
Risk Scoring and Security Dashboard

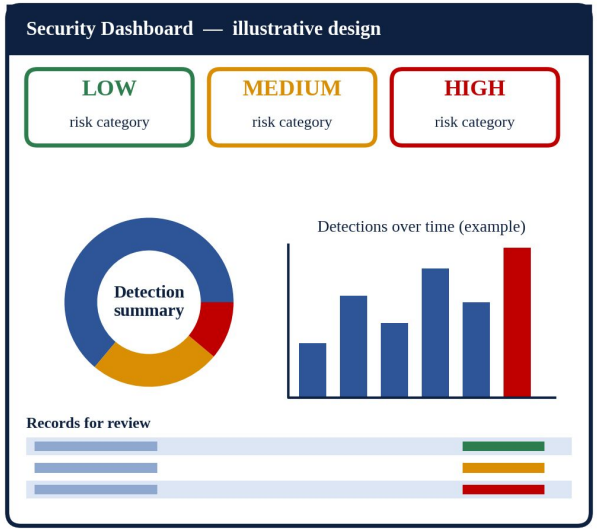
Purpose: Present detection results in an interpretable format to help users review potential threats.

Functions

- Receive outputs of spam and URL detection.
- Assign or display risk levels using documented rules or model outputs.
- Show contributing indicators where available.
- Display detection counts and category summaries.
- Present alerts, risk categories and records for review.
- Show evaluation metrics or trends when data exists.

Input: Predictions, supported confidence values, extracted indicators and available records.

Output: Risk categories or scores, dashboard summaries, and alerts or reports where implemented.



Illustrative interface design; not a screenshot of an implemented system.

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Module-wise Inputs and Outputs

Module	Input	Processing	Output
Module 1	Message and URL datasets	Data cleaning and feature extraction	Prepared data and feature representations
Module 2	Text records and relevant features	Spam classification	Spam / non-spam prediction
Module 3	URLs and relevant features	Malicious URL analysis	URL detection result
Module 4	Available detection results and supported indicators	Risk assignment and dashboard aggregation	Risk categories, summaries and alerts

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Technologies, Tools, Hardware and Software Requirements (Proposed)

Category	Technology / Requirement (proposed)
Programming language	Python
Data and text processing	Pandas, NumPy; NLTK or scikit-learn text utilities
Machine-learning framework	scikit-learn
Spam classification	Text features with a suitable classifier (to be finalized)
URL feature extraction	Python URL parsing and lexical feature extraction
Dashboard / user interface	Python-based dashboard framework (e.g., Streamlit)
Data storage	Local CSV files; SQLite if records must be stored
Development environment	VS Code / Jupyter Notebook
Operating system	Windows 10/11
Hardware	Laptop/PC, 8 GB RAM recommended

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Future Scope

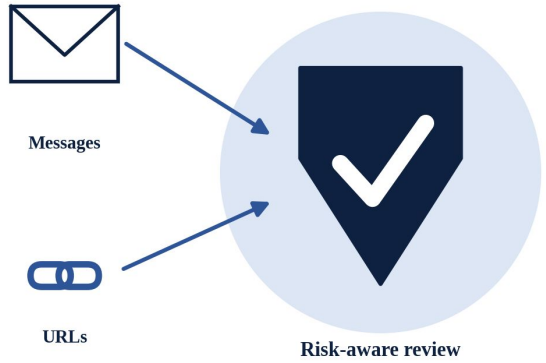
- Improve detection performance with **broader, well-labelled datasets**.
- Evaluate **alternative machine-learning or deep-learning models**.
- Improve handling of **previously unseen spam patterns** and suspicious URLs.
- Integrate **trusted threat-intelligence sources** if appropriate and available.
- Improve dashboard **filtering, reporting and explainability**.
- Add **periodic evaluation and model monitoring**.
- Investigate **real-time processing and alerting** as future work.

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Conclusion

A unified, risk-aware approach to detecting spam and malicious URLs.

- Identifying spam and suspicious URLs is essential for safer online communication.
- Dataset preparation and feature extraction provide the foundation for detection.
- Spam classification and malicious URL detection analyze their respective inputs.
- Risk scoring and a unified dashboard make results easier to review.
- Reliable test data and metrics are needed to evaluate the system.



References

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2025

Thank You!